$y = x^2 - 3$

$\hookrightarrow f(x) = x^2 - 3$

A function of one variable, $y = x^2 - 3$ or $f(x) = x^2 - 3$, has one input and one output. This unit extends that idea to functions with two or more inputs.

$f(x,y) = x^2 - \ln(y)$

$z = x^2 - \ln(y)$

$f(y,z) = x$

$f(x,z) = y$

$f(w,x,y,z)$

A function of two variables takes an input pair: $f(x, y) = x^2 - \ln y$, also written $z = x^2 - \ln y$, and its graph is a surface in space. The same idea works for $f(y, z)$, $f(x, z)$ or four inputs $f(w, x, y, z)$.

$f(x,y) = x^2 + y^2$

$x \in \mathbb{R} \quad y \in \mathbb{R}$

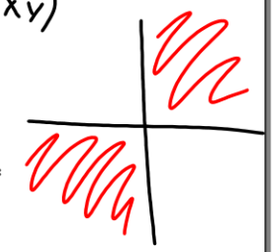
$\{(x,y) \mid x,y \in \mathbb{R}\}$

Domain = every input pair that gives a real output. $f(x, y) = x^2 + y^2$ has no restrictions, so the domain is all (x, y) with $x, y \in \mathbb{R}$: the whole plane.

$f(x,y) = \ln(xy)$

$D: xy > 0$

$\{(x,y) \mid xy > 0\}$



Domain of $\ln(xy)$: a log needs a positive input, so $xy > 0$. That means x and y have the same sign: the open first and third quadrants (shaded), with the axes left out.

$f(x,y) = \frac{\sqrt{x^2+y^2-9}}{x}$

$\{(x,y) \mid x^2+y^2 \geq 9, x \neq 0\}$



Two rules at once: the square root needs $x^2 + y^2 - 9 \geq 0$ (on or outside the circle of radius 3) and the fraction needs $x \neq 0$. Domain: on or outside the circle, minus the y-axis.

$f(x,y) = \sqrt{16 - x^2 - y^2}$

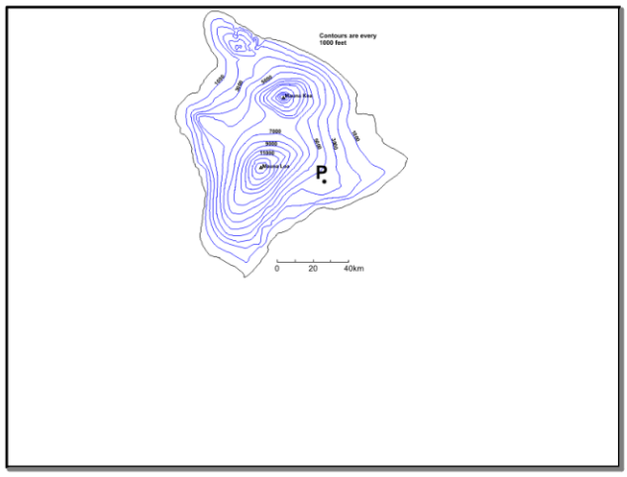
$D: \{(x,y) \mid x^2+y^2 \leq 16\}$

$R: \{z \mid 0 \leq z \leq 4\}$

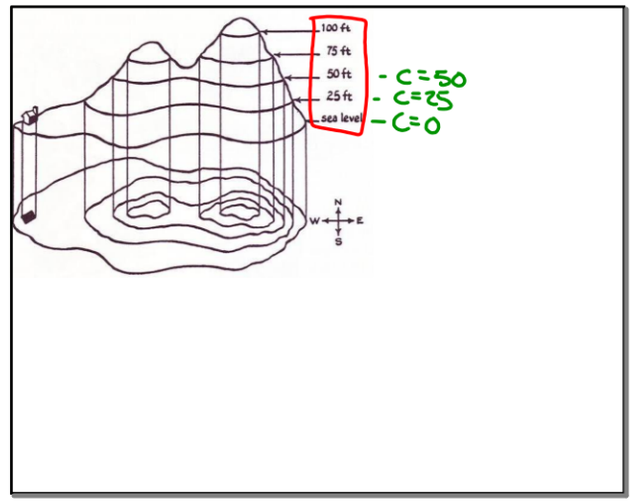


$\sqrt{(16 - x^2 - y^2)}$ needs $16 - x^2 - y^2 \geq 0$, so the domain is the disk $x^2 + y^2 \leq 16$. The output runs from 0 on the edge to $\sqrt{16} = 4$ at the origin, so the range is $0 \leq z \leq 4$. The graph is the top half of a sphere.

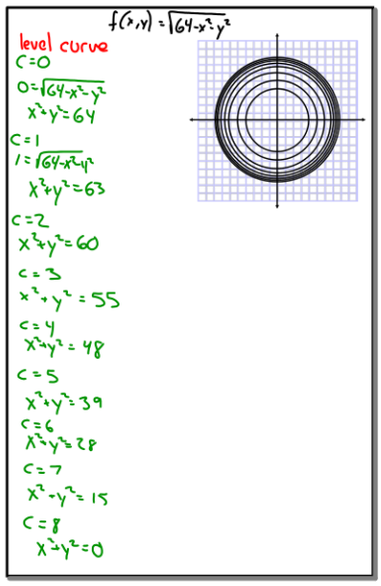
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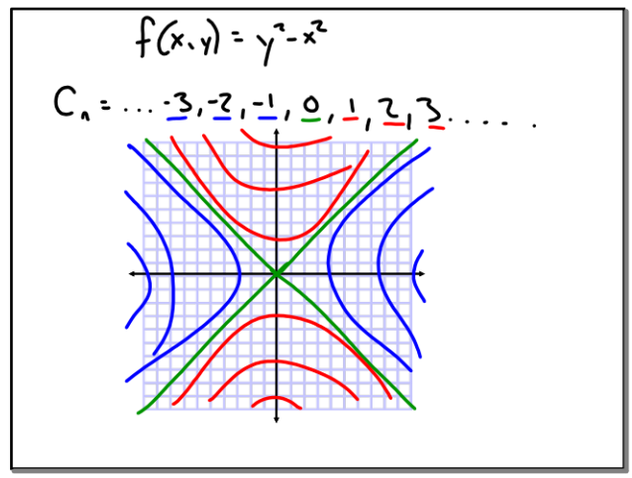
A contour (topographic) map: each curve joins points at the same height. Curves packed close together mean steep ground; curves far apart mean a gentle slope.



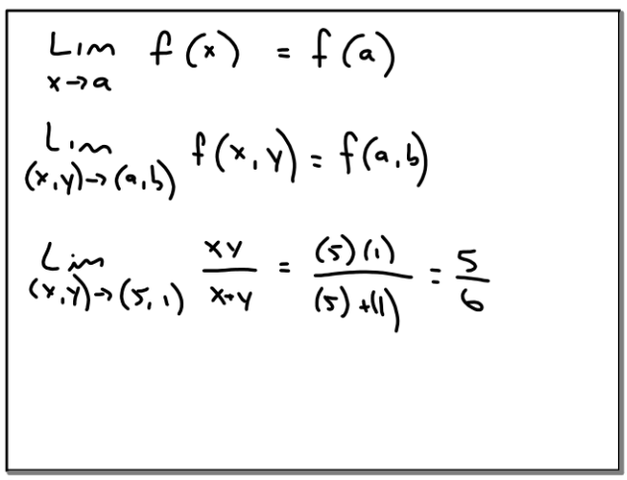
Where contour lines come from: slice the surface with horizontal planes z = c (here c = 0, 25, 50, ...) and drop each slice onto the xy-plane. Those curves f(x, y) = c are the level curves.



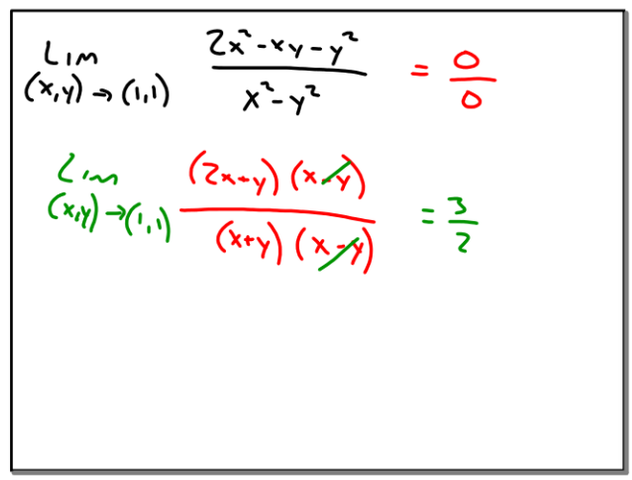
Level curves of f = √(64 − x² − y²): set f = c and square to get x² + y² = 64 − c². c = 0 gives radius 8, c = 4 gives √48, c = 7 gives √15, and c = 8 is just the origin. The circles crowd near the edge, where the hemisphere is steepest.



f = y² − x²: setting y² − x² = c gives hyperbolas, opening up/down for c > 0 and left/right for c < 0. c = 0 gives the crossing lines y = ±x. Together they map out a saddle.

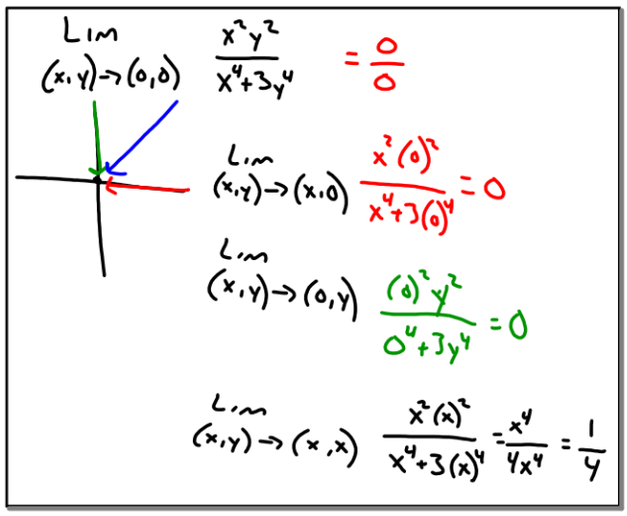


If f is continuous at (a, b), the limit is just f(a, b): plug the point in. lim (x,y)→(5,1) of xy/(x + y) = (5)(1)/(5 + 1) = 5/6, since nothing breaks at (5, 1).

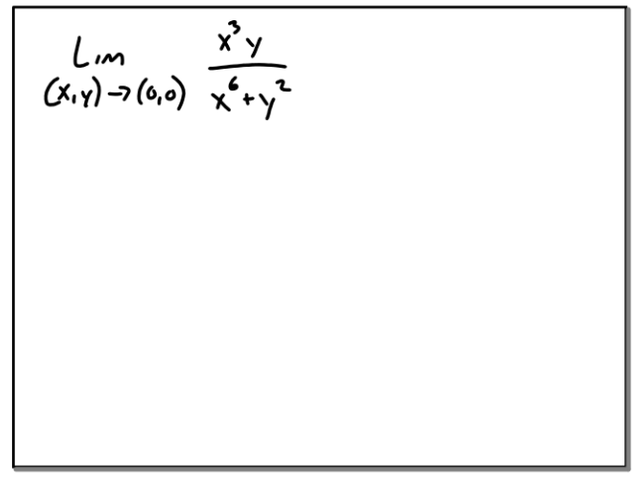


Plugging in gives 0/0, so simplify first. Factor 2x² − xy − y² = (2x + y)(x − y) and x² − y² = (x + y)(x − y), cancel x − y, then plug in: (2 + 1)/(1 + 1) = 3/2.

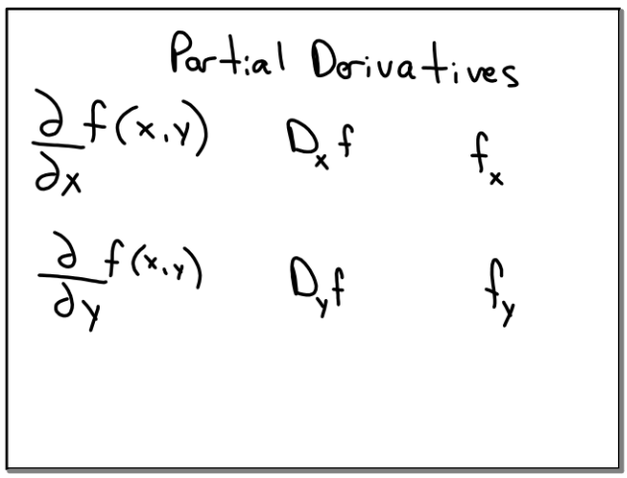
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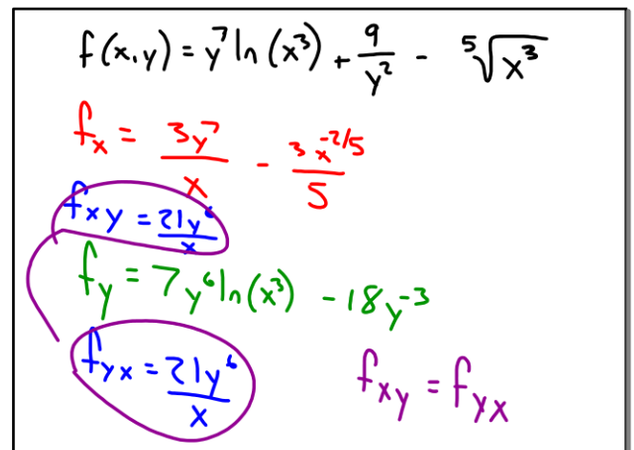
To show a limit at (0, 0) does not exist, find two paths that disagree. Along the x-axis ($y = 0$) and the y-axis ($x = 0$) the value is 0, but along $y = x$ it is $x^4/(4x^4) = 1/4$. Different answers, so the limit does not exist.



Not worked on the slide; use the same path test. Every line $y = mx$ gives 0, but along the curve $y = x^3$ the expression is $x^6/(2x^6) = 1/2$. Since $0 \neq 1/2$, the limit does not exist. Tip: pick a path that makes the powers match.



Partial derivatives: $\partial f/\partial x = D_x f = f_x$ differentiates in x while holding y constant; $\partial f/\partial y = D_y f = f_y$ holds x constant. Three notations for the same thing.



Treat the other variable as a constant. Since $\ln(x^3) = 3 \ln x$: $f_x = 3y^7/x - (3/5)x^{(-2/5)}$ and $f_y = 7y^6 \ln(x^3) - 18y^{-3}$. Both mixed partials come out $21y^6/x$, so $f_{xy} = f_{yx}$.